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Analyzing foundation models for segmentation of osseous metastatic lesions in prostate cancer on CT scans

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ABSTRACT

In metastatic prostate adenocarcinoma, accurate identification and quantification of metastatic lesions from CT images are essential for monitoring disease progression and evaluating therapeutic response. This study conducts a comparative evaluation of supervised learning and foundation model approaches for lesion segmentation, focusing on the zero-shot performance in a minimal computation setting. This comparison reflects real-world scenarios, where supervised learning is feasible when labeled data is available, whereas foundation models provide a flexible alternative in the absence of labeled. Furthermore, we investigate strategies to integrate supervised learning with foundation model methods. We first assess the performance of nnUNetv2, a state-of-the-art supervised segmentation framework, which achieved a Dice Similarity Coefficient (DSC) of 0.73 (95 % CI: 0.68–0.78). This serves as the baseline for comparison against three segmentation foundation models SaM-Med2D, SAM, and SAM2, when prompted using domain expert annotations (DEP) via bounding box. In the integrated approach, we examine nnUNetv2-generated prompts (nP) and domain expert prompts (DEP), along with their combinatorial operations with the baseline. While nP-based prompting combinations, such as SAM (DSC: 0.74 (95 % CI: 0.66–0.82; p = 0.06)) and SAM2 (DSC: 0.74 (95 % CI: 0.64–0.84; p = 0.07)), showed improved mean performance, none achieved statistically significant gains over the nnUNetv2 baseline. These findings highlight the robustness of supervised approaches like nnUNetv2 and suggest that while foundation models offer flexibility, their integration through prompting may require more adaptive strategies to yield tangible improvements in medical image segmentation.

1. Introduction

Prostate cancer (PCa) is the most prevalent and the second deadliest non-dermatologic cancer in men. According to the American Cancer Society's estimates for 2025, approximately 313,780 new cases of prostate cancer are expected in the United States, with about 35,770 deaths attributed to the disease. Notably, 1 in 8 men will be diagnosed with PCa during their lifetime [1,2]. PCa arises from the dysregulated proliferation of prostate epithelial cells, leading to malignant transformation and disease progression [3]. While localized PCa has a favorable prognosis, its progression to metastatic prostate cancer significantly worsens outcomes [4]. The poor prognosis of metastatic

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disease is due to widespread involvement that precludes complete surgical resection, the accumulation of somatic mutations in tumor cells that propagate cancer development, compression of vital structures from tumor volume, and disruption of molecular pathways. The clinical outcomes are particularly poor with the development of bony metastases which is classically characterized by aberrant bone formation and osteoblastic lesions. (Fig. 1) [5–7]. The skeletal system is the most common metastatic site, with over 80 % of advanced PCa cases exhibiting bone involvement, primarily affecting the spine, pelvis, and ribs. Bone metastases lead to severe complications such as pathological fractures, spinal cord compression, and bone pain, reducing the quality of life and survival rates. Given the aggressive nature of metastatic PCa, early detection, qualification, quantification of osseous metastatic lesions in medical imaging, such as MRI, CT, PET, SPECT, and bone scintigraphy are essential for timely intervention [8]. However, PCa identification through imaging poses several challenges, including variability in shape, size, heterogeneity of tissue appearance, ambiguity of tissue boundaries, and inter-observer variability [9]. To address these challenges, AI-based methods driven by deep learning, machine learning and radiomics techniques have been leveraged to enhance the precision, consistency, and automation of lesion detection and segmentation [10, 11].

To contextualize our study within the existing body of work, we summarize key recent advances in AI-driven segmentation of bone metastases below. Hinzpeter et al. [12] conducted a proof-of-concept study using radiomics of computed tomography (CT) imaging data to differentiate bone metastases, including those not visible on CT, from normal bone tissue, with ^{68}Ga-PSMA PET imaging as the reference standard. The model (gradient boosted tree) demonstrated a classification accuracy of 0.85 (CI: 0.76–0.92, $p < .001$), with a sensitivity of 78 % and specificity of 93 %, highlighting its strong diagnostic performance. However, this study is limited by its reliance on ^{68}Ga-PSMA PET as the reference standard, which may not be universally available. A retrospective radiomics study by Acar et al. [13] explored machine learning for differentiating metastatic and treatment-responded sclerotic bone lesions in prostate cancer patients based on texture features. The findings suggest that CT imaging with texture analysis and machine learning (weighted KNN) can effectively distinguish these lesions, achieving an AUC of 0.76. Reischauer et al. [14] assessed whole-lesion volumetric texture analysis of apparent diffusion coefficient (ADC) maps to evaluate treatment response in prostate cancer bone metastases. Texture analysis before and after androgen deprivation therapy (ADT) revealed that most features exhibited moderate to high precision and underwent significant changes post-treatment. Additionally, several first-order and second-order statistical features showed strong correlations with serum PSA levels, highlighting their potential for treatment monitoring. However, the study relied on ADC maps, which may not be routinely acquired in all clinical settings, affecting generalizability. A study by Wang et al. [15] explored the use of multiparametric MRI-based texture analysis to predict bone metastases in prostate cancer patients before treatment. The researchers extracted 976 texture features from T2-weighted and dynamic contrast-enhanced T1-weighted MRI scans of 176 patients. They found that a combination of 15 selected features significantly correlated with the presence of bone metastases. The predictive model, which integrated these MRI-derived features with clinical risk factors such as PSA levels and Gleason scores, achieved an area under the curve (AUC) of 0.916, indicating high diagnostic performance.

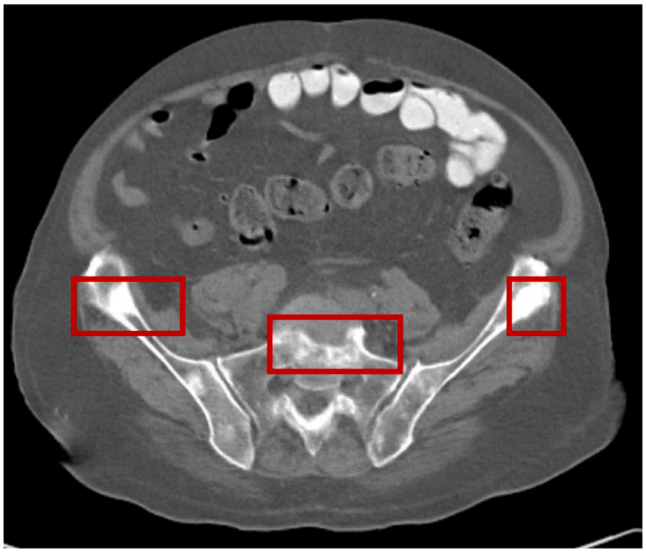


Fig. 1. Axial CT scan in a 74-year patient with multiple sclerotic osteoblastic metastasis in the pelvic bones.

A study by Anderson et al. [16] developed machine learning models to estimate survival durations in patients with bone metastases from prostate cancer. Analyzing records of 438 men treated for skeletal-related events, the models predicted 1- to 10-year survival with acceptable calibration, accuracy (Brier scores < 0.20), and classification ability (AUCs > 0.73). Key factors positively associated with longer survival included younger age at metastasis diagnosis, proximal prostate-specific antigen (PSA) levels below 10 ng/mL, and slower alkaline phosphatase velocity. A study by Liu et al. [17] developed and validated a machine learning model to predict bone metastases in prostate cancer patients. Utilizing data from the SEER database (2010–2017), they analyzed demographic and clinicopathologic variables of 207,137 patients, identifying factors such as Gleason score, PSA value, T stage, N stage, and age as significant risk factors. Among six machine learning algorithms tested, the eXtreme Gradient Boosting (XGB) model demonstrated the best performance, with an AUC of 0.962.

Afnouch et al. [18] introduced BM-Seg, a comprehensive dataset designed to capture the diversity of bone metastases in terms of location, organ involvement, and severity from CT scans. They also proposed an ensemble of CNN-based segmentation approaches to effectively segment metastatic lesions achieving a Dice Similarity Coefficient (DSC) score of 0.77, indicating promising performance. Although the study does not perform patient-wise stratification, which may lead to data leakage and may overestimate performance. In [19], the authors validated multiple deep learning models for detecting and segmenting osseous metastatic prostate cancer lesions on CT scans. Efficient Net demonstrated superior detection performance (F1 score: 0.82, AUC: 0.71), while nnUNetv2 outperformed others in segmentation (DSC: 0.74). Additionally, radiomic analysis confirmed that deep learning-based segmentations closely matched manual segmented lesions, reinforcing their clinical applicability. Masoudi at al. presented a cascaded deep learning algorithm for automated detection and classification of bone lesions in prostate cancer staging CT scans. The method consists of a two-stage framework: (1) a detection algorithm based on YOLO-v3 with a darknet-53 backbone, achieving 81 % sensitivity and 86 % positive predictive value, and (2) a classification model using ResNet-50, which distinguished benign from malignant lesions with 89 % accuracy. Belue et al. [20] developed an AI ensemble for bone lesion detection, segmentation, and classification in prostate cancer (PCa) staging CTs, aiming to improve efficiency and reduce reliance on expert analysis. Using 297 CT scans (4601 benign and 1911 metastatic lesions), two AI models were trained: 3DAISeg for lesion segmentation and 3DAIClass for benign vs. metastatic classification. In 45 unseen scans, 3DAISeg detected 73.1 % of metastatic and 72.4 % of benign lesions, achieving a Dice score of 0.53 (median: 0.59, range: 0–0.87). 3DAIClass achieved an F1-score of 94.8 % (radiologist contours) and 92.4 % (AI contours), with positive and negative predictive values (PPV/NPV) comparable to radiologists. The AI model performed on par with junior and senior radiologists, showing potential for assisting in prostate cancer bone metastasis diagnosis.

While the deep learning models have significantly improved detection and segmentation performance, they still face limitations such as

domain generalization, and variations in lesion appearance. Foundation models, characterized by pretraining on diverse, large-scale datasets, offer potential advantages in enhancing segmentation accuracy and efficiency. These models leverage extensive pretraining to generalize better across datasets, potentially overcoming the limitations of traditional AI models that rely heavily on annotated training data. However, their clinical applicability is still under evaluation, as challenges related to robustness, interpretability, and computational efficiency remain. In this study, we aim to compare the performance of supervised deep learning (nnUNetv2) with foundation segmentation models for prostate cancer bone metastasis segmentation. Beyond this comparison, we investigate whether prompting strategies using nnUNetv2-generated outputs or expert annotations can enhance the performance of foundation models. Our goal is to assess the potential of these integrated approaches to improve segmentation accuracy, reduce inter-observer variability, and support clinical decision-making in quantifying bone metastases in prostate cancer patients.

2. Methods

The dataset consists of 1925 lesions from 8863 2D images obtained from 23 male patients diagnosed with metastatic prostate carcinoma, enrolled in a clinical trial at our institution. The mean age was 70.0 $\pm$ 10.2 years (range 45–93). Among the study cohort, 43 % of patients were on Abiraterone, 35 % were on Cabazitaxel, and 22 % were on Docetaxel. This information has been added to the Methods in the revised version of the manuscript. All data collection and processing procedures were conducted under Institutional Review Board (IRB) approval and are fully compliant with the Health Insurance Portability and Accountability Act (HIPAA), ensuring the ethical handling and confidentiality. To enhance the focus on the bone region, pixel intensity values were normalized by adjusting Hounsfield Units (HU) to a fixed range, improving the consistency and reliability of the imaging data for subsequent analysis.

All scans were acquired using Toshiba systems following standardized chest and abdomen/pelvis protocols. For chest imaging, scans were obtained from above the lung apices to the mid-kidney level (120 kVp, 110–500 mA, slice thickness 2 mm, interval 2 mm, rotation time 0.35 s, pitch = 0.813). Dose modulation and a standard body reconstruction kernel (FC08) were applied. Intravenous contrast was administered at a rate of 1.5–2.5 mL/s. For abdomen/pelvis imaging, scans extended from above the diaphragm to the pubic symphysis (120 kVp, 70–500 mA, slice thickness 2 mm, interval 2 mm, rotation time 0.5 s, pitch = 0.813). Dose modulation was enabled, and images were reconstructed using a standard body kernel (FC08). Portal venous phase images were acquired 70 s post-contrast injection.

The annotation process was conducted using the Synapse PACS under the guidance of a senior radiologist with over 20 years of experience in radiology and oncology imaging. The annotations were performed by a radiologist with over 10 years of clinical experience, ensuring high-quality and accurate labeling. While the involvement of two experienced radiologists ensured high-quality and clinically informed labeling, we acknowledge that no formal intra- or inter-observer variability assessment was performed in this study. Binary annotation was employed, with each pixel classified as either positive (1) or negative (0) metastatic lesion to effectively distinguish lesion regions. Throughout the development, we employed a hold-out approach with an 80:20 split, allocating 80 % of the data for training and 20 % for testing, while ensuring that no subject was represented in more than one split. Furthermore, we used paired *t*-tests to compare the Dice Similarity Coefficient (DSC) scores between models and applied a False Discovery Rate (FDR) correction using the Benjamini–Hochberg procedure, maintaining an overall significance threshold of $\alpha = 0.05$ to determine statistical significance.

We utilize nnUNetv2 [21] as the baseline architecture, a self-configuring deep learning framework. Unlike conventional methods that require extensive manual tuning, nnUNetv2 automatically optimizes preprocessing, network architecture, and hyperparameters based on dataset characteristics. Our goal is two-fold: first, to compare the performance of this supervised method against foundation models such as SAMMed2D [22], SAM [23], and SAM2 [23], which leverage advanced architectures and large-scale pretraining for better generalization across diverse medical imaging scenarios. These foundation models offer greater flexibility by requiring only prompts, such as points, bounding boxes, instead of segmentation masks, which are more time-consuming to create. Second, we investigate the integration of supervised and foundation models through prompting strategies, aiming to enhance performance beyond that of the supervised baseline. In this regard, we incorporate two types of prompts: (1) nnUNetv2-generated prompts (nP) and (2) domain expert prompts (DEP), which provide clinically relevant guidance. For nnUNetv2-generated prompts (nP), each lesion predicted by nnUNetv2, a single point was automatically generated inside the predicted mask. Specifically, the geometric centroid of the connected component was used. For very small predicted lesions, the pixel closest to the lesion center was chosen. If nnUNetv2 failed to generate a mask for a lesion slice, no nP was supplied. In DEP, for each lesion and axial slice containing that lesion, an experienced radiologist drew a single axis-aligned bounding box tightly enclosing the lesion, with a small fixed margin to ensure full coverage. When multiple lesions appeared on the same slice, separate boxes were drawn. If a lesion was not visible on a slice, no DEP was provided for that slice. These boxes were used directly as prompts for SAM, SAM2, and SAMMed2D models. Additionally, we evaluate combinatorial strategies, such as the union and intersection between the masks generated by nnUNetv2 and those generated by foundation models to assess potential performance improvements. In the union ($\cup$), a region was labeled positive if predicted by either model, whereas in the intersection ($\cap$), a region was labeled positive only if predicted by both. While union captured more of the metastatic extent, it also introduced false positives, and both strategies tended to underestimate lesion boundaries. Fig. 2 illustrates the overall workflow of the study.

3. Results

Table 1 presents the performance of various segmentation methods measured by the Dice Similarity Coefficient (DSC). The supervised nnUNetv2 model (M0) achieved a DSC of 0.73 (95 % CI: 0.68–0.78), serving as the baseline. On the other hand, foundation models prompted with domain expert input (DEP) via bounding boxes (BB) were not as accurate with SAM, SAM2 and SamMed2D achieving DSCs of 0.70 (95 % CI: 0.63–0.77; $p = 0.07$), 0.70 (95 % CI: 0.66–0.74; <0.01), and 0.68 (95 % CI: 0.62–0.74; $p < 0.01$), respectively. Overall, nnUNetv2 was more accurate when compared to prompt-based foundation models.

Thereafter, we integrated the baseline (M0) with foundation models. We evaluated the performance of the union of the masks generated by M0 and M1, M2 and M3. The combinations M1-U-M0, M2-U-M0, and M3-U-M0 achieved DSC scores of 0.75 (95 % CI: 0.69–0.81; $p = 0.31$), 0.72 (95 % CI: 0.65–0.79; $p = 0.34$), and 0.71 (95 % CI: 0.63–0.79; $p = 0.18$), respectively. Similarly, the intersection of the masks M1-$\cap$-M0, M2-$\cap$-M0, and M3-$\cap$-M0 achieved DSC scores of 0.71 (95 % CI: 0.64–0.78; $p = 0.18$), 0.71 (95 % CI: 0.63–0.79; $p = 0.19$), and 0.74 (95 % CI: 0.68–0.80; $p = 0.59$), respectively. While the combinations M1-U-M0 and M3-$\cap$-M0 performed slightly better than the baseline, the improvement was not statistically significant.

The foundation models were also tested with prompts generated by the nnUNet model. These models, denoted as M4, M5, and M6, achieved DSCs of 0.74 (95 % CI: 0.67–0.81; $p = 0.57$), 0.73 (95 % CI: 0.67–0.79; $p = 0.57$), and 0.71 (95 % CI: 0.65–0.77; $p = 0.03$), corresponding to SAM, SAM2, and SAMMed2D, respectively.

The performance of SAM was better than the baseline nnUnetv2 model, though this gain was not statistically significant. The union of the masks generated by the nnUnetv2 model and the foundation models

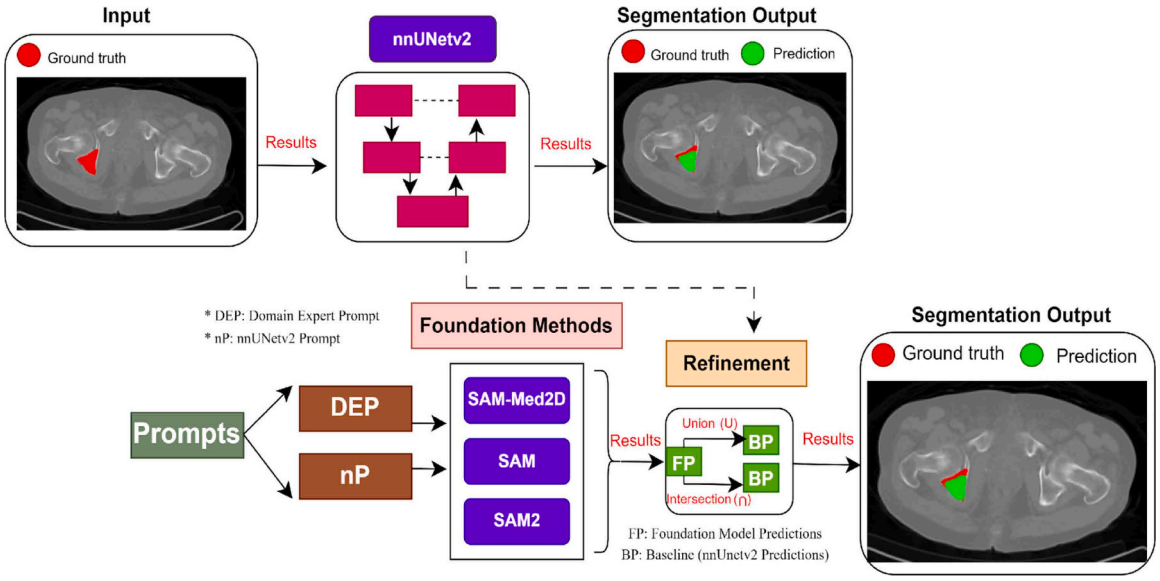


Fig. 2. Overview of the methodology: The baseline nnUNetv2 architecture performs initial segmentation. Foundation models (SAM, SAM2, SamMed2D) are integrated using Domain Expert Prompts (DEP) and nnUNetv2-based prompts (nP), with combinatorial strategies applied to refine and enhance segmentation performance. (Note: The red label refers to the ground truth as segmented by radiologists and the green refers to the predicted lesion.).

Table 1
Comparison of segmentation performance between supervised vs foundation methods.

Model label	Label required	Prompt source	Segmentation model	DSC
M0	Mask	None	nnUNetv2	0.73 (95 % CI: 0.68–0.78)
M1	BB	DEP	SAM	0.70 (95 % CI: 0.63–0.77; p = 0.07)
M2	BB	DEP	SAM2	0.70 (95 % CI: 0.66–0.74; <0.01)
M3	BB	DEP	SAMMed2D	0.68 (95 % CI: 0.62–0.74; p < 0.01)
M4	Mask	nnUNetv2	SAM	0.74 (95 % CI: 0.67–0.81; p = 0.57)
M5	Mask	nnUNetv2	SAM2	0.73 (95 % CI: 0.67–0.79; p = 0.57)
M6	Mask	nnUNetv2	SAMMed2D	0.71 (95 % CI: 0.65–0.77; p = 0.03)
M1-∪-M0	Mask+BB	DEP	nnUNetv2 ∪ SAM	0.75 (95 % CI: 0.69–0.81; p = 0.31)
M1-∩ -M0	Mask+BB	DEP	nnUNetv2 ∩ SAM	0.71 (95 % CI: 0.63–0.79; p = 0.18)
M2-∪-M0	Mask+BB	DEP	nnUNetv2 ∪ SAM2	0.72 (95 % CI: 0.65–0.79; p = 0.34)
M2-∩ -M0	Mask+BB	DEP	nnUNetv2 ∩ SAM2	0.71 (95 % CI: 0.63–0.79; p = 0.19)
M3-∪-M0	Mask+BB	DEP	nnUNetv2 ∪ SAMMed2D	0.71 (95 % CI: 0.64–0.78; p = 0.18)
M3-∩ -M0	Mask+BB	DEP	nnUNetv2 ∩ SAMMed2D	0.74 (95 % CI: 0.68–0.80; p = 0.59)
M4-∪-M0	Mask	nnUNetv2	nnUNetv2 ∪ SAM	0.74 (95 % CI: 0.66–0.82; p = 0.06)
M4-∩ -M0	Mask	nnUNetv2	nnUNetv2 ∩ SAM	0.73 (95 % CI: 0.65–0.81; p = 0.05)
M5-∪-M0	Mask	nnUNetv2	nnUNetv2 ∪ SAM2	0.74(95 % CI: 0.64–0.84; p = 0.07)
M5-∩ -M0	Mask	nnUNetv2	nnUNetv2 ∩ SAM2	0.72 (95 % CI: 0.66–0.78; <0.01)
M6-∪-M0	Mask	nnUNetv2	nnUNetv2 ∪ SAMMed2D	0.72 (95 % CI: 0.67–0.77; p = 0.03)
M6-∩ -M0	Mask	nnUNetv2	nnUNetv2 ∩ SAMMed2D	0.71 (95 % CI: 0.63–0.79; p = 0.03)

denoted by M4-U-M0, M5-U-M0, and M6-U-M0, achieved DSC scores of 0.74 (95 % CI: 0.66–0.82; p = 0.06), 0.74(95 % CI: 0.64–0.84; p = 0.07), and 0.72 (95 % CI: 0.67–0.77; p = 0.03), respectively. Similarly, the intersection of these masks, denoted by M4-∩ -M0, M5-∩ -M0, and M6-∩ -M0 achieved DSC scores of 0.73 (95 % CI: 0.65–0.81; p = 0.05), 0.72 (95 % CI: 0.66–0.78; <0.01), and 0.71 (95 % CI: 0.63–0.79; p = 0.03), respectively. Of these, only M4-U-M0 and M5-U-M0 performed better than the baseline, and this improvement was close to be statistically significant.

4. Discussion

This study investigated the comparative performance of supervised learning models and foundation models in segmenting metastatic lesions from CT images and further investigated strategies for integrating their results.

When compared with supervised learning models, which require entire masks for training, foundation models only need prompts such as points or bounding boxes to guide segmentation. This reduces the requirement on labeled data and is a definite advantage of these models. However, in our tests we observed that this advantage comes at the cost of lower accuracy as all the foundation models (M1, M2, M3) that used DEP prompts were not as accurate as the baseline nnUNetv2 model trained using segmentation masks.

Not surprisingly, we also observed that the prompt used in a foundation model can have a significant impact on its performance. This can be observed by comparing the performance of foundation models that used DEP prompts (M1, M2, and M3) with those that used nnUnetv2 prompts (M4, M5, and M6). In each case, for the same foundation model we observed a significant change in performance by just changing the prompt type.

Finally, we considered whether a combination of the masks produced

by the supervised learning model and foundation models could be used to produce more accurate results. For this we considered the union and intersection of these masks, and observed that, in most cases, masks obtained by considering the union were more accurate. However, none of the masks (either those based on the union or the intersection) were more accurate (with statistical significance) than the baseline supervised learning mask. The fact that the union of the masks was more accurate than intersection points to the fact that the masks generated by nnU-Netv2 and the foundation models, all likely underestimated the spatial extent of the lesions. Thus, considering their union, rather than intersection generated more accurate results. It is worth noting the lower performance of foundation models compared to the supervised baseline (nnUNetv2), which can be attributed to a few key factors. First, a domain shift exists between the data used to pretrain foundation models and the target domain of medical imaging. Models such as SAM and SAM2 were pretrained on large-scale natural image datasets and were not specifically exposed to grayscale CT scans or oncologic imaging features. This mismatch likely limited their ability to capture subtle patterns in Hounsfield Units, fine-grained anatomical detail, and lesion-specific contrast variations relevant to osseous metastases. Second, lesion heterogeneity presents an added challenge. Bone metastases can vary widely in their size, shape, density, and spatial distribution, making them difficult to detect and segment without task-specific training. In contrast, the supervised nnUNetv2 model, trained directly on lesion-level annotations from the same imaging modality, is better equipped to learn these complex patterns, contributing to its stronger overall performance.

This study has certain limitations that we want to be acknowledged. The dataset was collected from a single institution, which may influence the generalizability of the results across different clinical settings. Although annotations were performed under the supervision of a senior radiologist and carried out by an experienced radiologist, no formal intra- or inter-observer variability assessment was conducted. Additionally, foundation models were evaluated in their pretrained form without fine-tuning; future work could explore whether task-specific fine-tuning further improves performance. Lastly, while we explored basic composite strategies using union and intersection operations to integrate supervised and foundation model outputs. Future work could focus on more adaptive or learning-based fusion strategies to fully realize the potential of hybrid segmentation pipelines.

5. Conclusion

In this study, we first compared the segmentation performance of a supervised model (nnUNetv2) with foundation models (SAMMed2D, SAM, and SAM2) and observed that the supervised approach was more accurate. Foundation models, when guided solely by expert prompts (DEP), underperformed relative to the baseline, indicating limited standalone utility in the context of prostate bone metastasis segmentation. We then explored prompting strategies as a means of integrating supervised and foundation models. nnUNet-based prompts (nP) offered more consistent guidance, resulting in modest performance gains for some models. We also considered masks obtained from the union and intersection of the masks generated by the supervised learning model and foundation models. Once again, we observed that none of the combinations were significantly more accurate that the supervised learning baseline. Overall, in the context of prostate bone metastasis segmentation, we conclude that the use of foundation models does not lead to more accurate results. However, despite their reduced accuracy, they may still be an attractive option in a low-resourced setting, where masks required from training a supervised model are not available, and simple bounding-box type prompts can be easily generated.

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Disclosures

VD has Consultancy engagements with Radmetrix, Roche, Deeptek, and Westat.

Institutional Review Board Statement

This study protocol was reviewed and approved by the University of Southern California Institutional Review Board (IRB), approval number HS-036031.

Ethical Statement

This retrospective study was approved by the Institutional Review Board (approval number HS-036031) of the University of Southern California, Keck School of Medicine. The requirement for informed consent was waived due to the use of de-identified data, and all procedures were conducted in compliance with the Health Insurance Portability and Accountability Act (HIPAA) regulations.

CRediT authorship contribution statement

Joseph Rich: Writing – original draft, Visualization, Data curation. **Ian Smith:** Validation, Data curation. **Vinay Duddalwar:** Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Inga Van Buren:** Validation, Data curation. **Assad A Oberai:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Shreyas Malewar:** Writing – original draft, Methodology. **Passant Mohamed:** Data curation. **Pawan S J:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Amir Goldkorn:** Supervision. **Timothy Triche:** Supervision. **Steven Y Cen:** Writing – original draft, Validation, Formal analysis, Data curation. **Xiaomeng Lei:** Formal analysis, Data curation. **Darryl H Hwang:** Validation, Supervision, Data curation. **R Prajwal:** Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

1. Joseph Rich reports grant support from the Radiological Society of North America.
2. Vinay Duddalwar reports grant support from the Ming Hsieh Foundation, Samsung Healthcare, National Institutes of Health, Wright Foundation and Mann Foundation. He also reports consulting or advisory relationships with Radmetrix, Westat Inc., Roche, and DeepTek.
3. Amir Goldkorn reports grant support from the National Institutes of Health and the Wright Foundation.

All other authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

The data that has been used is confidential.

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